

Exercises

1	2	3	4	5	6	7	8	9	10
11									

Surname, First name**5EZB0 Math 2**

Interim test Q4

1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9
0	0	0	0	0	0	0

☐ a	☒ b	☐ c	☐ d	☐ e	☐ f	→ b
☒ a	☐ b	☐ c	☒ d	☐ e	☐ f	→ a
☒ a	☐ b	☒ c	☐ d	☐ e	☐ f	→ c
☒ a	☐ b	☒ c	☒ d	☐ e	☐ f	→ d

Answer multiple-choice questions as shown in the example.

Particular Ans on paper exam instructions

- Write in a black or blue pen.
- The exam has multiple-choice questions. Select the correct answer by marking it as shown above.
- Hand in all pages. Do not remove the staple. If you remove it anyhow, check that you hand in all pages.

Dear student,

You're about to take an exam. Write down your name and your student ID at the appropriate places above. Make sure that you enter your student ID by fully coloring the appropriate boxes. On the examination attendance card, you fill in the PDF number. You can find the correct number on the top of the first page of your exam (e.g. 1234.pdf).

Please read the following information carefully:

Date exam: 22/05/2025

Start time 15:45

End time: 17:15

Number of questions: 11

Maximum number of points: 50

Method of determining the final grade: (Points obtained)/5

Answering style: multiple choice.



Permitted examination aids

- Scrap paper (fully blank)
- Calculator(**graphic/ basic**)

Important:

- You are only permitted to visit the toilets under supervision
- Examination scripts (fully completed examination paper, stating name, student number, etc.) must always be handed in
- The house rules must be observed during the examination
- The instructions of subject experts and invigilators must be followed
- Keep your workplace as clean as possible: put pencil case and breadbox away, limit snacks and drinks
- You are not permitted to share examination aids or lend them to each other
- Do not communicate with any other person by any means

During written examinations, the following actions will in any case be deemed to constitute fraud or attempted fraud:

- using another person's proof of identity/campus card (student identity card)
- having a mobile telephone or any other type of media-carrying device on your desk or in your clothes
- using, or attempting to use, unauthorized resources and aids, such as the internet, a mobile telephone, smartwatch, smart glasses etc.
- having any paper at hand other than that provided by TU/e, unless stated otherwise
- copying (in any form)
- visiting the toilet (or going outside) without permission or supervision

You can start the exam now, good luck!

Multivariable functions

3p **1** Where are the local extrema of the function $f(x, y) = x^2 + y^2 - 4y + 9$ located?

- ☐ a The only critical point is located at $(2, 0)$ and is a saddle point.
- ☐ b The only minimum is located at $(0, 2)$.
- ☐ c The criteria are not sufficient.
- ☐ d The minimum is located at $(0, 2)$ and the maximum is at $(0, -2)$
- ☐ e There are no local extrema.

Coordinate systems

3p **2a** What is the region

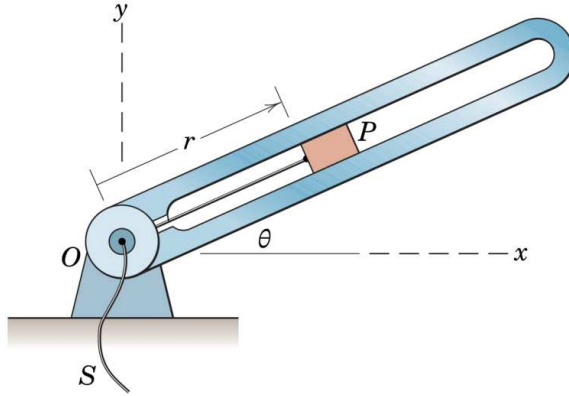
$$R \leq 2 \cos(\phi)$$

in spherical coordinates?

- ☐ a a saddle
- ☐ b a paraboloid
- ☐ c a sphere
- ☐ d a cone
- ☐ e a cylinder

4p 2b

Consider a slider P in a slotted arm, as depicted in the figure below. The slider can be moved inward by string S , while the slotted arm rotates around point O . The angular position θ of the arm is prescribed as $\theta = 0.8t - 0.05t^2$ rad, where t indicates time. The position of slider P inside the slot can be described as a function of time as $r = 1.6 - 0.2t$ m.



The motion of slider P is best described in cylindrical coordinates. What is the velocity $\mathbf{v}$ of slider P at $t = 1$ s?

- (a) $\mathbf{v} = -0.2 \hat{\mathbf{r}} + 0.75 \hat{\boldsymbol{\theta}}$
- (b) $\mathbf{v} = -0.2 \hat{\mathbf{r}} + 0.98 \hat{\boldsymbol{\theta}}$
- (c) $\mathbf{v} = 1.4 \hat{\mathbf{r}} + 0.98 \hat{\boldsymbol{\theta}}$
- (d) $\mathbf{v} = -0.2 \hat{\mathbf{r}}$
- (e) $\mathbf{v} = -0.2 \hat{\mathbf{r}} + 0.7 \hat{\boldsymbol{\theta}}$
- (f) $\mathbf{v} = 1.4 \hat{\mathbf{r}} + 0.75 \hat{\boldsymbol{\theta}}$
- (g) $\mathbf{v} = 1.4 \hat{\mathbf{r}}$
- (h) $\mathbf{v} = 1.4 \hat{\mathbf{r}} + 0.7 \hat{\boldsymbol{\theta}}$

Parametrization

4p 3 The intersection between the plane $x = 1$ and the paraboloid $z = x^2 + y^2$ is a curve. What is the equation of the tangent to the curve at the point $P = (1, 2, 5)$?

- (a) $y = 2x, z + 2y = 9$
- (b) $z = 5, x + 2y = 3$
- (c) $\frac{x-1}{2} = \frac{y-2}{4} = z - 5$
- (d) $x = 1 + t, y = 2 + t, z = 5 + 4t$
- (e) $x = 1, z = 4y - 3$

Limits

4p 4 Consider the function $f(x, y) = \frac{x \sin(y)}{\sqrt{x^2 + y^2}}$. The limit $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$

- (a) Depends on the direction of approach
- (b) is 1
- (c) is $+\infty$
- (d) does not exist
- (e) is 0

Arc length

4p 5 An ant goes around the spiral path $(t \cos t, t \sin t, \frac{1}{6}t^3)$ for all $t > 0$. What is the length it traverses in the first half turn, i.e., between $t = 0$ and $t = \pi$.

- (a) $\pi + \frac{\pi^3}{6}$
- (b) $\frac{\pi^4}{27} + \sqrt{2} - 1$
- (c) $\sqrt{4\pi^2 + \frac{1}{36}\pi^6}$
- (d) $\int_0^\pi \sqrt{2 + t^2} dt$

Domain of definition

4p 6 Consider the function $f(x, y) = 1 - \sqrt{xy + x^2}$. Let $\mathcal{D} \subset \mathbb{R}^2$ be its maximum domain of definition. Choose the true statement

- (a) The domain $\mathcal{D}$ is open, and $(0, 0)$ is not an isolated point of the domain
- (b) The domain $\mathcal{D}$ is closed, and $(0, 0)$ is not an isolated point of the domain
- (c) The domain $\mathcal{D}$ is neither open nor closed, and $(0, 0)$ is an isolated point
- (d) The domain $\mathcal{D}$ is closed, and $(0, 0)$ is an isolated point of the domain
- (e) The domain $\mathcal{D}$ is open, and $(0, 0)$ is an isolated point of the domain

Partial derivatives

4p 7 For a constant $m > 0$, consider the function

$$f(x, y) = \begin{cases} 0 & \text{if } (x, y) = (0, 0), \\ \frac{\sin(|x|^m)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0). \end{cases}$$

Its y -partial derivative $\partial_y f$ is continuous at $(0, 0)$

- (a) For $m > 3$
- (b) for no $m > 0$
- (c) For $m \leq 2$
- (d) For $m > 2$
- (e) For $m \leq 4$

Chain rule

- 4p **8** Consider the topography whose height is given by

$$H(x, y) = \sin(\pi xy^2)/\pi$$

for $x, y \in \mathbb{R}$. You are walking on a road such that your x and y coordinates are changing with time t as

$$x(t) = t^3 + 1, \quad y(t) = 1 - 2t^2.$$

What is your rate of ascent ($\partial_t H(x(t), y(t))$) and horizontal speed v (speed in xy -plane) at time $t = 1$?

- ☐ a $\partial_t H = -13, v = 5$
- ☐ b $\partial_t H = -19, v = -5$
- ☐ c $\partial_t H = 13, v = 5$
- ☐ d $\partial_t H = 13, v = -5$
- ☐ e $\partial_t H = 19, v = 5$

Directional derivatives

- 4p **9** Temperature profile in $\mathbb{R}^3$ is given by

$$T(x, y, z) = (x^2 + 3y)e^{-z}.$$

You are standing at $(1, 1, 0)$. In which direction should you throw a ball so that it experiences the fastest temperature drop?

- ☐ a $(-2, -3, 4)$
- ☐ b $(2, 3, -4)$
- ☐ c $(-3, -2, -3)$
- ☐ d $(3, 2, 3)$

Higher order derivatives and Taylor series

- 4p **10a** Consider the function $f(x, y) = \sin(1 + xy)$. What is the determinant of the Hessian matrix $D^2 f$ at $(x, y) = (0, 0)$?
- (a) $-\sin(1) \cos(1)$
 - (b) $\cos(1)^2$
 - (c) $-\cos(1)^2$
 - (d) $-\sin(1)^2$
- 4p **10b** What is the Taylor series approximation of the function f (as given in **10a**) up to the second degree around the point $(0, 0)$?
- (a) $\sin(1) + \cos(1)xy$
 - (b) $\sin(1) - \frac{\sin(1)}{2}(x + y)^2$
 - (c) $\sin(1) + \cos(1)(x + y) - \frac{\sin(1)}{2}(x^2 + y^2)$
 - (d) $\sin(1) + \cos(1)(x + y) - \sin(1)xy$

Conservative forces

4p 11 Consider a potential energy function U that is a depends on Cartesian x and y coordinates as follows:

$$U = \sin(\pi x) \sin(2\pi y)$$

The conservative force $\mathbf{F}$ that relates to U at point $(x, y) = (\frac{1}{3}, \frac{1}{3})$ is:

- (a) $\mathbf{F} = -\frac{1}{4}\pi\sqrt{3}\hat{i} - \frac{1}{2}\pi\sqrt{3}\hat{j}$
- (b) $\mathbf{F} = -\frac{3}{4}\pi\hat{i} + \frac{3}{2}\pi\hat{j}$
- (c) $\mathbf{F} = \frac{3}{4}\pi\hat{i} + \frac{3}{2}\pi\hat{j}$
- (d) $\mathbf{F} = \frac{1}{4}\pi\sqrt{3}\hat{i} - \frac{1}{2}\pi\sqrt{3}\hat{j}$
- (e) $\mathbf{F} = -\frac{3}{4}\pi\hat{i} - \frac{3}{2}\pi\hat{j}$
- (f) $\mathbf{F} = -\frac{1}{4}\pi\sqrt{3}\hat{i} + \frac{1}{2}\pi\sqrt{3}\hat{j}$
- (g) $\mathbf{F} = \frac{1}{4}\pi\sqrt{3}\hat{i} + \frac{1}{2}\pi\sqrt{3}\hat{j}$
- (h) $\mathbf{F} = \frac{3}{4}\pi\hat{i} - \frac{3}{2}\pi\hat{j}$

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